

C++ Assignment: Custom Exceptions and Stream Operator Overloading

Domain: Banking / Digital Wallet System

Problem Statement

Develop a C++ application to manage a simple Digital Wallet system. The system should store wallet holder details and wallet balance. The application must demonstrate the use of custom exception handling and operator overloading for the insertion (<<) and extraction (>>) operators.

Class Requirements

- Create a class `Wallet` with the following data members:
 - `walletId`
 - `holderName`
 - `balance`

Provide suitable constructors and member functions to initialize and display wallet details.

Custom Exception Requirements

- Create a custom exception class `InvalidAmountException`.
- Throw the exception when the user enters an invalid transaction amount, such as zero or a negative value.
- Create another custom exception class `InsufficientBalanceException`.
- Throw the exception when a withdrawal amount is greater than the available wallet balance.
- Handle all exceptions using appropriate try-catch blocks and display meaningful error messages.

Operator Overloading Requirements

- Overload the >> operator to accept `Wallet` details from the user.
- Overload the << operator to display `Wallet` details in a formatted manner.
- Implement these operators as friend functions.

Functional Requirements

1. Accept wallet details using the overloaded >> operator.
2. Display wallet details using the overloaded << operator.
3. Provide options to add money and withdraw money.
4. Validate transaction amounts using custom exceptions.
5. Display the updated wallet information after every successful transaction.
6. Use a menu-driven program to perform multiple operations until the user chooses to exit.

Expected Learning Outcomes

- Understand and implement user-defined/custom exception classes.
- Use throw and try-catch blocks for exception handling.
- Overload the stream insertion operator (<<).
- Overload the stream extraction operator (>>).
- Understand the use of friend functions in operator overloading.
- Design a menu-driven object-oriented C++ application.

Sample Menu

1. Create Wallet
2. Add Money
3. Withdraw Money
4. Display Wallet Details
5. Exit